

Image corpus + feature extractor

Semantic-health gates (development split)

Gate 1: Weighted-feature responsiveness

*Every nonzero-weight
feature must vary*

Gate 2: Retained absolute formula weight

*Post-fix requirement:
 $\sum |w| = 1.0$*

Gate 3: Rank-equivalent baseline rejection

*Reject $|\rho| = 1$
duplicates*

Gate 4: Exported-score boundary-mass audit

*Clipping, ties,
unique-value count*

Held-out evaluation (G-test)

*Fail-fast
on violation*

